THE EXACT MAXIMUM WEAK CHROMATIC VILLAINY AT FIXED GRAPH ORDER AND CHROMATIC NUMBER

JIANG JIANG, YUHAN JIANG, YIYANG LI, LUJIA WANG, AITAO ZHANG

Editorial note. Complete original author mathematical body, all core and supporting proofs and embedded bibliography retained; one independent main conclusion per task. No proof repair, added hypotheses or independent correctness certification.

Editorial note. Mechanical source0.tex adapter omits only pdfTeX-specific glyphtounicode input and pdfgentounicode switch, which are unnecessary under XeLaTeX. Original raw TeX retained; mathematics unchanged.

Allowed standard prerequisites (editorial).

- Proper colorings, Hamming repair distance and integer transportation/assignment duality. Native ordinary/weak repair distinctions, feasible profiles, target multiplicities, dominant block parameters and complete supporting quota/assignment arguments retained.. All author core constructions remain present; in particular the nonadjacent prescribed-quota proof is included explicitly as part of the odd-cycle theorem proof bundle.

1. Introduction

A rearrangement of the colors of a proper coloring preserves the number of occurrences of each color but may destroy properness. Chromatic villainy measures the greatest distance from such a rearrangement to a proper coloring with the same color multiplicities. More precisely, if G has chromatic number k, its villainy $B(G)$ is the maximum, over all proper k-colorings and all rearrangements of their colors, of the minimum number of vertices that must change color in a count-preserving repair. Its weak villainy $B_{\mathrm{w}}(G)$ allows the repaired coloring to have different multiplicities, while using the same palette of k colors. Both distances count changed vertices. They concern the initial and final colorings, with no prescribed sequence of intermediate recolorings.

Clark et al. [5] introduced these parameters and determined them for several basic graph classes, including connected bipartite graphs. For odd cycles they proved $B_{\mathrm{w}}(C_{2r+1}) = r$ and $B(C_{2r+1}) \geq r$ for $r \geq 2$, and asked whether the latter bound is always attained. They also raised questions about complete multipartite graphs and the largest possible villainy at fixed order and chromatic number. These questions appear in West's collection [17] and in the account of Jahanbekam and Lin [10], who characterized the graphs with villainy two. Raleigh's thesis [16] obtained further exact values and bounds for complete multipartite graphs. It also treated two odd-cycle profiles [16, Theorems 7 and 8]. The general odd-cycle question remained open in both that thesis and the later paper of Jahanbekam and Lin.

We approach these problems through the *profile* of a coloring (Theorem 2.1), the vector of its labeled color multiplicities. For a feasible profile $\mathbf{m}$, let $R_G(\mathbf{m})$ be the largest, over all color assignments of profile $\mathbf{m}$, of the distance to a proper coloring of that same profile. Then $B(G) = \max_{\mathbf{m}} R_G(\mathbf{m})$, with the maximum over feasible profiles. This viewpoint keeps track of the restriction that distinguishes ordinary villainy from weak villainy, and allows the admissible source and target profiles to be varied separately. On cycles, this leads to exact agreement guarantees at prescribed profiles. On complete multipartite graphs, the same count constraints give an assignment problem. Completing a graph relative to an optimal coloring then extends the multipartite description to the largest possible values over all graphs of given order and chromatic number.

Our first result determines the full directed repair radius on an odd cycle. For any source profile $\mathbf{m}$ and any feasible target profile $\mathbf{p}$, Theorem 3.15 gives

$$R_{C_{2r+1}}(\mathbf{m} \to \mathbf{p}) = 2r + 1 - \sum_{X} \min\{p_X, \lceil m_X/2 \rceil\}.$$

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The source profile need not itself be feasible. In fact, Theorem 3.3 guarantees at least $\min\{p_X, \lceil m_X/2 \rceil\}$ agreements separately for each color X. A source word with each color confined to one consecutive block gives the radius for every target profile simultaneously, so Theorem 3.15 also treats any nonempty family of allowed target profiles.

Taking the same feasible profile $\mathbf{m} = (a, b, c)$ at source and target yields

$$R_{C_{2r+1}}(\mathbf{m}) = \left\lfloor \frac{a}{2} \right\rfloor + \left\lfloor \frac{b}{2} \right\rfloor + \left\lfloor \frac{c}{2} \right\rfloor,$$

and hence $B(C_{2r+1}) = r$ for $r \geq 2$, as stated in Theorem 3.1. The matching obstruction and the floor sum already occur in the weak-repair argument of Clark et al. [5, Proposition 3.5]; the upper bound here preserves every labeled multiplicity. The ordinary and weak radii therefore agree at every feasible source profile. The proof constructs auxiliary targets and interpolates for proper sources, then uses induction for arbitrary sources.

For a complete multipartite graph $K_{p_1,\dots,p_k}$, a disturbance is described by a nonnegative integer matrix whose row and column sums are $\mathbf{p}$. A weak repair chooses a maximum-weight assignment. An ordinary repair restricts the assignment to parts of equal size. The relevant groups of parts are therefore the *size classes*, each consisting of all parts of one size. The maximal-value case is known: $B(G) = |V(G)|$ exactly when no size class contains more than half the vertices. Raleigh proved the sufficient direction [16, Theorem 1]; the equivalence also follows from Cameron and Wanless's covering-radius theorem for permutation groups [4, Theorem 14]. Call a size class *dominant* if it contains more than half the vertices. An additional dominant-class regime was determined by Raleigh [16, Theorem 4].

Theorem 4.3 gives a general arithmetic formula, including the remaining dominant-class cases, with at most k^2 candidates. For weak villainy, Theorem 4.7 expresses the answer as a finite minimum over ordered integral assignment potentials. A strengthened exchange bounds their coordinate gaps by differences of part sizes, up to one. Thus the explicit search depends on the size spread rather than a common baseline. Theorem 4.9 also gives fixed-parameter tractability in k for binary-encoded sizes through a fixed-dimensional integer program: the running time is $f(k)L^c$, where L is the input length, f is computable, and c is an absolute constant. When all part sizes are equal, the assignment problem is the integer doubly stochastic problem treated by Dinitz, Hartman, and Soprunova [6]; Gajula, Soprunov, and Soprunova [9] studied its rectangular extension. The formulas here permit arbitrary positive part sizes. Integral assignment duality and transportation cuts provide a common framework for the two repair rules, although their resulting optimizations are different.

We also determine both extremal values

$$\mathcal{M}(n,k) = \max_{\substack{|V(G)|=n \\ \chi(G)=k}} B(G), \qquad \mathcal{M}_{\mathrm{w}}(n,k) = \max_{\substack{|V(G)|=n \\ \chi(G)=k}} B_{\mathrm{w}}(G).$$

Both maxima are attained by complete multipartite graphs. The reduction completes a graph relative to a proper coloring whose profile supplies an extremal disturbance; the disturbance remains admissible after completion. For $k \geq 3$, the ordinary extremum is $\min\{n, 4(n-k)\}$ apart from the explicitly listed exceptions in Theorem 5.2. For weak villainy, Theorem 5.3 gives, for $k \geq 2$, $\mathcal{M}_{\mathrm{w}}(n,k) = n - L - \varepsilon$, where $L = \max\{\lceil n/k \rceil, 2k-n\}$ and the correction $\varepsilon \in \{0,1\}$ is determined by a parity obstruction. The obstruction comes from balance of the integer disturbance matrix: its corresponding row and column sums agree. Explicit balanced matrices show that this is the only correction required; attaining weak extremizers need at most three distinct part sizes.

The majority-retention theorem is related to coloring with requests [7]: each vertex requests one color, and one seeks a proper coloring satisfying many requests. Quantitative request-satisfaction bounds and structural aspects of flexible list coloring are studied in [12, 2]. Here the quotas are separate for each requested color, and the target multiplicities are prescribed simultaneously. Prescribed color-class sizes also arise in the sampling results of Kuchukova, Perkins, and Povill [13] and the recent existence theorem of Birken [1].

Section 2 gives the profile formulation and the transportation criterion. Section 3 proves the odd-cycle results. Sections 4 and 5 treat complete multipartite graphs and the extrema at fixed order and chromatic number. We conclude with questions about extremal structure, arithmetic complexity, and prescribed-profile retention.

2. Color profiles and a transportation criterion

All graphs are finite and simple. Unless otherwise stated, a graph G is nonempty, has order n, and has chromatic number k. We use a fixed set $\mathcal{C}$ of k labeled colors. For a positive integer s, write $[s] = \{1, \dots, s\}$.

Definition 2.1 (Words and profiles)**.** A *word* on G is a map $w : V(G) \to \mathcal{C}$, with no properness assumption. Its *profile* is the vector $\mathbf{m}(w) = (m_X(w))_{X \in \mathcal{C}}$, where $m_X(w) = |w^{-1}(X)|$. Bold letters denote vectors. Inequalities between vectors are coordinatewise, $\mathbf{e}_X$ is a coordinate unit vector, and $\mathbf{1}$ is an all-one vector of the required dimension.

Write $\mathcal{P}(G)$ for the proper colorings with palette $\mathcal{C}$. Since $|\mathcal{C}| = \chi(G)$, each such coloring uses every color. A profile $\mathbf{m}$ is *feasible* if it occurs in $\mathcal{P}(G)$.

For a feasible profile, put

$$\mathcal{W}_{\mathbf{m}}(G) = \{w : V(G) \to \mathcal{C} : \mathbf{m}(w) = \mathbf{m}\}, \qquad \mathcal{P}_{\mathbf{m}}(G) = \mathcal{W}_{\mathbf{m}}(G) \cap \mathcal{P}(G).$$

We omit G from $\mathcal{W}_{\mathbf{m}}(G)$ when the vertex set is clear. Every word in $\mathcal{W}_{\mathbf{m}}(G)$ is a rearrangement of every proper coloring with that profile; we call it a *legal disturbance*. More generally, we call the given word the *source* and a proper coloring used to repair it the *target*. An *agreement* is a vertex on which source and target have the same color. The Hamming distance and the number of agreements are

$$d_{\mathrm{H}}(w, y) = |\{v : w(v) \neq y(v)\}|, \qquad \mathrm{agr}(w, y) = n - d_{\mathrm{H}}(w, y).$$

The cost therefore counts changed vertices; exchanging two different colors on two vertices has cost two.

Definition 2.2 (Repair radii and villainy)**.** The repair radius of a profile and its weak counterpart are

$$R_G(\mathbf{m}) = \max_{w \in \mathcal{W}_{\mathbf{m}}(G)} \min_{y \in \mathcal{P}_{\mathbf{m}}(G)} d_{\mathrm{H}}(w, y),$$
$$R_G^{\mathrm{w}}(\mathbf{m}) = \max_{w \in \mathcal{W}_{\mathbf{m}}(G)} \min_{y \in \mathcal{P}(G)} d_{\mathrm{H}}(w, y).$$

With the terminology of Clark et al. [5],

$$B(G) = \max_{\mathbf{m} \text{ feasible}} R_G(\mathbf{m}), \qquad B_{\mathrm{w}}(G) = \max_{\mathbf{m} \text{ feasible}} R_G^{\mathrm{w}}(\mathbf{m}).$$

Both parameters are zero on the empty graph by convention.

We also call $B(G)$ *ordinary villainy*, to distinguish it from weak villainy.

The inclusion of the two target sets gives $B_{\mathrm{w}}(G) \leq B(G)$. In particular, the weak target may change the profile, but it still uses the same palette of $\chi(G)$ colors. A simultaneous permutation of the color labels in the source and target preserves distances and all profile conditions.

We use the following integral transportation criterion, a classical consequence of the max-flow theorem [8]. Its short proof is included because the precise capacity inequalities are used both for gap allocations on cycles and for multipartite graphs.

For a vector $\mathbf{a}$ indexed by a finite set, write $a(I) = \sum_{i \in I} a_i$.

Lemma 2.3 (Integral transportation criterion)**.** *Let $\mathbf{a} \in \mathbb{Z}_{\geq 0}^h$ and $\mathbf{b} \in \mathbb{Z}_{\geq 0}^j$ have the same total N, and let $c_{i\ell}$ be nonnegative integers. There is a nonnegative integer matrix M with row sums $\mathbf{a}$, column sums $\mathbf{b}$, and $m_{i\ell} \leq c_{i\ell}$ if and only if*

$$a(I) + b(J) - N \leq \sum_{i \in I, \ell \in J} c_{i\ell} \qquad (I \subseteq [h],\ J \subseteq [j]). \tag{2.1}$$

Proof. At most $N - b(J)$ units from rows in I can lie outside the columns in J. This proves necessity. For sufficiency, form a network whose arcs from the source to row i, from row i to column ℓ, and from column ℓ to the sink have capacities $a_i, c_{i\ell}, b_\ell$, respectively. A cut with source-side rows I and sink-side columns J has capacity

$$2N - a(I) - b(J) + \sum_{i \in I, \ell \in J} c_{i\ell}.$$

The assumed inequalities make every cut have capacity at least N. Starting with zero flow, augment along residual source–sink paths by positive integers. The value increases at every step and is bounded by N. If augmentation stops before value N, the vertices reachable from the source determine a cut whose capacity is the current value, a contradiction. An integral flow of value N saturates all source and sink arcs. Its row-to-column values give the required matrix. □

The same augmenting-path argument applies to every integral network used below. In particular, when the row and column bounds are upper bounds rather than prescribed sums, the largest possible integral total is the minimum cut capacity.

3. Odd cycles and prescribed color multiplicities

Let $n = 2r + 1$, and use the three labeled colors $\mathcal{C} = \{A, B, C\}$. We identify a coloring of C_n with a cyclic word w. Its profile is $\mathbf{m} = (m_A, m_B, m_C)$, where $m_X = |w^{-1}(X)|$. The profile is *feasible* if some proper coloring has these multiplicities. We write X^a for a consecutive block of a copies of color X, with the block empty when $a = 0$.

Theorem 3.1. *For every feasible profile* $\mathbf{m} = (a, b, c)$ *of* C_{2r+1},

$$R_{C_{2r+1}}(\mathbf{m}) = \left\lfloor \frac{a}{2} \right\rfloor + \left\lfloor \frac{b}{2} \right\rfloor + \left\lfloor \frac{c}{2} \right\rfloor = \begin{cases} r - 1, & a, b, c \text{ are all odd,} \\ r, & \text{exactly one of } a, b, c \text{ is odd.} \end{cases} \tag{3.1}$$

Consequently, $B(C_{2r+1}) = r$ *for* $r \geq 2$, *and* $B(C_3) = 0$.

Clark et al. [5, Proposition 3.5 and Corollary 3.6] proved $B_{\mathrm{w}}(C_{2r+1}) = r$ and $B(C_{2r+1}) \geq r$ for $r \geq 2$. The floor sum in (3.1) also occurs in their argument. Raleigh [16, Theorems 7–8] proved upper bounds for the profiles $(r, r, 1)$ and $(r, r - 1, 2)$, respectively. Her first proof already gives $2\lfloor r/2 \rfloor$, and the second gives r; these agree with (3.1). They are profile-restricted statements, not the full odd-cycle formula. The issue addressed here is the requirement that the repaired coloring preserve each labeled color multiplicity. We prove a stronger assertion in which the target multiplicities may be prescribed independently of the source multiplicities.

3.1. Profiles and agreement quotas.

Lemma 3.2. *A nonnegative integer vector* $\mathbf{m}$ *is feasible on* C_{2r+1} *if and only if*

$$\sum_{X \in \mathcal{C}} m_X = 2r + 1, \qquad 1 \leq m_X \leq r \quad (X \in \mathcal{C}). \tag{3.2}$$

Proof. Each color class of a proper coloring is independent and therefore has at most r vertices. All three colors occur, since an odd cycle is not bipartite. This proves necessity.

Conversely, suppose (3.2) holds. Form a loopless multigraph on A, B, C in which the multiplicity of the edge AB is $m_A + m_B - m_C$, and define the other two multiplicities cyclically. For example, the multiplicity of AB is $2r + 1 - 2m_C \geq 1$. Thus this multigraph is connected, has $2r + 1$ edges, and has degree $2m_X$ at X. It has an Euler circuit. Reading the successive vertices of this circuit, with the final repeated vertex omitted, gives a cyclic word in which consecutive letters differ and X occurs m_X times. □

For an arbitrary source word w, define its half-count requirements by

$$\mathbf{q} = (q_A, q_B, q_C), \qquad q_X = \left\lceil \frac{m_X}{2} \right\rceil.$$

The source need not be proper, and some of its counts may be zero. For a feasible target profile $\mathbf{p}$, the corresponding agreement quotas are

$$\mathbf{t} = (t_A, t_B, t_C), \qquad t_X = \min\{p_X, q_X\}.$$

The target count caps a color's half-count requirement when fewer vertices of that color are available in the target.

Theorem 3.3 (Prescribed multiplicities and capped majority retention)**.** *Let* w *be a word on* C_{2r+1}, *where* $r \geq 1$, *with profile* $\mathbf{m}$. *If an integer vector* $\mathbf{p}$ *satisfies*

$$\sum_{X \in \mathcal{C}} p_X = 2r + 1, \qquad 1 \leq p_X \leq r \quad (X \in \mathcal{C}), \tag{3.3}$$

then there is a proper coloring y *of profile* $\mathbf{p}$ *such that*

$$|\{v : w_v = y_v = X\}| \geq t_X = \min\{p_X, q_X\} \quad (X \in \mathcal{C}). \tag{3.4}$$

For distinct colors X, Y, we shall use the inequalities

$$q_X + q_Y \le r + 1 \le p_X + p_Y. \tag{3.5}$$

The first follows from $m_X + m_Y \le 2r + 1$, and the second from the upper bound r on the third target count. In particular, if $p_A < q_A$, both other target counts strictly exceed their half-count requirements. We call a target coordinate X *deficient* if $p_X < q_X$ and *slack* if $p_X > q_X$. At most one coordinate can be deficient.

We first deduce the profile-radius formula, assuming this theorem.

Proof of Theorem 3.1. For a feasible profile $\mathbf{m}$, set $\mathbf{p} = \mathbf{m}$ in Theorem 3.3. Its hypotheses hold by Theorem 3.2, and then $t_X = q_X$ for every color. Every source word of profile $\mathbf{m}$ has a proper repair of the same profile retaining at least $\sum_X \lceil m_X/2 \rceil$ vertices. Hence

$$R_{C_{2r+1}}(\mathbf{m}) \le \sum_{X \in \mathcal{C}} \left\lfloor \frac{m_X}{2} \right\rfloor.$$

The matching lower bound is the three-block obstruction in the proof of Clark et al. [5, Proposition 3.5]: for the source $A^a B^b C^c$, every proper coloring differs on at least $\lfloor a/2 \rfloor + \lfloor b/2 \rfloor + \lfloor c/2 \rfloor$ vertices, even when target multiplicities are unrestricted. It applies in particular to the smaller family of proper targets of profile $\mathbf{m}$, proving equality.

If $o \in \{1, 3\}$ is the number of odd entries of $\mathbf{m}$, then

$$\sum_X \left\lfloor \frac{m_X}{2} \right\rfloor = \frac{2r + 1 - o}{2}.$$

This gives the two cases in (3.1). The value is always at most r. For even $r \ge 2$, the feasible profile $(r, r, 1)$ attains r; for odd $r \ge 3$, the feasible profile $(r, r-1, 2)$ does so. When $r = 1$, the sole feasible profile is $(1, 1, 1)$, and every rearrangement is already proper. □

There are two parts to the proof of Theorem 3.3. We first treat proper sources by constructing suitable auxiliary profiles and interpolating while keeping a specified set of agreements fixed. We then remove an equal adjacent pair from an improper source and apply induction. Two elementary observations about alternating colorings will also be useful.

Lemma 3.4. *Suppose a set whose vertices are colored B or C is partitioned into E and O, where $\big||E| - |O|\big| \le 1$. Of the two assignments*

$$(B \text{ on } E,\ C \text{ on } O) \quad \text{and} \quad (C \text{ on } E,\ B \text{ on } O),$$

one retains at least the ceiling of half of the original vertices of each color.

Proof. Write b_E, b_O, c_E, c_O for the original color counts in the two parts. The first assignment retains b_E vertices of color B and c_O vertices of color C; the second retains b_O and c_E, respectively. Their agreement differences are

$$d_B = b_E - b_O, \qquad d_C = c_O - c_E.$$

Since $d_B - d_C = |E| - |O|$, the integers d_B, d_C cannot have opposite strict signs. Consequently one assignment is at least as good as the other for both colors. For each color, its two possible agreement counts sum to its original multiplicity, so the better count is at least the ceiling of half that multiplicity. □

To apply this alternating choice after deleting a vertex of the cycle, we need a root that balances the original vertices of one color between the two parity classes of the remaining path, namely the vertices in its odd and even positions.

Lemma 3.5. *For every nonempty vertex set S in an odd cycle, there is $s \in S$ such that the two parity classes of the path $C_n - s$ contain numbers of vertices of $S \setminus \{s\}$ differing by at most one.*

Proof. For $s \in S$, let $\delta(s)$ be the number of other vertices of S at even clockwise distance from s, minus the number at odd distance. If $|S| = 1$, then $\delta(s) = 0$ and there is nothing to prove. Otherwise let t be the next vertex of S clockwise from s, at distance g. No other vertex of S lies between them, so shifting the root from s to t gives

$$\delta(t) = \begin{cases} \delta(s) - 2, & g \text{ even}, \\ -\delta(s), & g \text{ odd}. \end{cases} \tag{3.6}$$

To see this, distances to all other vertices decrease by g. The vertex t, at distance g from the old root, is replaced in the count by s, at distance $n-g$ from the new root. Since n is odd, these two distances have opposite parity, giving (3.6).

Suppose $|\delta(s)| > 1$ for every $s \in S$. Across an even gap, subtracting two cannot change the sign without producing a value of absolute value at most one. Across an odd gap the sign reverses. The gaps around the cycle sum to odd n, so an odd number of them are odd. After one complete tour the sign would therefore reverse, although the root has returned to its starting point. This contradiction proves the existence of $|\delta(s)| \le 1$. □

Theorems 3.4 and 3.5 settle the case in which one target color occurs only once.

Lemma 3.6. *Theorem 3.3 holds if a target multiplicity is* 1.

The same-profile singleton case has the elementary alternating-path proof in [16, Theorem 7]. We include the following argument because the induction needs independent source and target profiles, possibly with zero source counts, and a separate agreement quota for each color.

Proof. Relabel the colors so that $p_C = 1$. The sum and upper bounds in (3.3) force $p_A = p_B = r$. If the source is monochromatic, every proper coloring of this target profile has exactly $p_X = t_X$ agreements in its sole source color X, so any such coloring suffices. Otherwise every source count is at most $2r$, and hence $q_A, q_B \le r$.

First suppose $m_C \ge 1$. By Theorem 3.5, choose an original C vertex s such that the two parity classes of $C_n - s$ contain numbers of other original C vertices differing by at most one. Keep s colored C. Each parity class has r vertices, so after ignoring the other original C positions, the remaining original A, B positions in the two classes also have sizes differing by at most one. Theorem 3.4, with the colors renamed, therefore selects an alternating A, B coloring retaining at least q_A, q_B agreements. Assign its alternating colors also to the ignored positions. The root is the unique target C, so its incident edges are proper, and its one agreement meets $t_C = 1$.

Now suppose $m_C = 0$. Of m_A, m_B, precisely one is even. Its value is positive because the source is not monochromatic. Choose a vertex of the positive even-count color and assign it the unique target C. Decreasing a positive even count by one leaves its half-ceiling unchanged. On the remaining path, whose two parity classes have equal size, Theorem 3.4 therefore gives an alternating coloring meeting the original A, B quotas. The C quota is zero. □

3.2. **Interpolation with fixed agreements.** We next describe the profiles of all proper extensions of a prescribed partial coloring: these are proper colorings that agree with every prescribed color. The unprescribed positions between consecutive prescribed vertices form a *gap*, which may be empty. The fact that there are exactly three colors is essential: the relevant allocation conditions reduce to one lower and one upper bound for each coordinate.

Lemma 3.7. *Consider a path with $\ell \ge 0$ internal vertices and prescribed endpoint colors s, t. If $\ell = 0$, assume $s \ne t$. Let e_X be the number of endpoints colored X, counted with multiplicity. An integer vector $\mathbf{x}$ is the profile of a proper filling of the internal vertices if and only if*

$$\sum_{X \in \mathcal{C}} x_X = \ell, \qquad 0 \le x_X \le \left\lfloor \frac{\ell + 1 - e_X}{2} \right\rfloor \quad (X \in \mathcal{C}). \tag{3.7}$$

Proof. Every internal occurrence of X is incident with two edges, and every endpoint occurrence with one. Properness prevents an edge from having both ends colored X. Since the path has $\ell + 1$ edges, $2x_X + e_X \le \ell + 1$, proving the upper bounds.

For sufficiency, form a loopless multigraph on the three colors. Give the edge joining the other two colors, opposite X, multiplicity

$$h_X = \ell + 1 - 2x_X - e_X.$$

The assumptions make these multiplicities nonnegative integers. Their sum is $\ell + 1$, and the degree at X is $2x_X + e_X$. A loopless multigraph on three vertices has at most one component containing edges. Both prescribed endpoint colors belong to that component. If $s \ne t$, precisely s and t have odd degree, so there is an Euler trail from s to t; if $s = t$, there is an Euler circuit starting there. The sequence of colors along this trail gives a proper path. Its number of internal occurrences of X is $(\deg(X) - e_X)/2 = x_X$, as required. □

We use the following specialization of the integral transportation criterion, Theorem 2.3. Suppose a gap g has ℓ_g positions and can receive at most u_{gX} occurrences of color X. Assume all these quantities are nonnegative integers and $\sum_X u_{gX} \geq \ell_g$. For nonnegative integer demands d_X of total $M = \sum_g \ell_g$, a filling of all gaps with those color totals exists if and only if

$$\sum_{X\in S} d_X \leq \sum_g \min\left\{\ell_g, \sum_{X\in S} u_{gX}\right\} \quad (S \subseteq \mathcal{C}). \tag{3.8}$$

Indeed, apply Theorem 2.3 with rows indexed by the gaps, row sums ℓ_g, column sums d_X, and capacities u_{gX}. For a fixed set S of columns, its inequalities for all row sets I are equivalent to

$$\sum_{X\in S} d_X \leq \sum_{g\notin I} \ell_g + \sum_{g\in I}\sum_{X\in S} u_{gX}.$$

Minimizing the right side independently for each row gives exactly (3.8).

We now combine the gap capacities with this allocation criterion to describe every profile compatible with a fixed partial coloring.

Lemma 3.8 (Fixed-set interpolation)**.** *Let $K \neq \varnothing$ be a set of vertices of C_n with prescribed colors admitting a proper extension. Suppose profiles $\mathbf{x}$ and $\mathbf{z}$ occur among its proper extensions. Then an integer vector $\mathbf{p}$ also occurs as the profile of a proper extension of K whenever*

$$\sum_X p_X = n, \qquad \min\{x_X, z_X\} \leq p_X \leq \max\{x_X, z_X\} \quad (X \in \mathcal{C}).$$

Proof. Let k_X be the number of prescribed vertices colored X. The vertices outside K split into gaps between consecutive vertices of K in cyclic order. Include gaps of length zero. If $|K| = 1$, the same fixed vertex supplies both endpoints of the single gap. Write ℓ_g for the number of internal vertices in gap g and e_{gX} for the number of its endpoints colored X.

Theorem 3.7 gives the capacities

$$u_{gX} = \left\lfloor \frac{\ell_g + 1 - e_{gX}}{2} \right\rfloor.$$

The assumed proper extension ensures that these capacities are nonnegative and that $\sum_X u_{gX} \geq \ell_g$ for every gap. Moreover, $u_{gX} \leq \ell_g$. Define

$$U_X = k_X + \sum_g u_{gX},$$

$$L_X = k_X + \sum_g \max\left\{0, \ell_g - \sum_{Y\neq X} u_{gY}\right\}.$$

We claim that the extension profiles are precisely

$$\left\{\mathbf{p} \in \mathbb{Z}^3 : \sum_X p_X = n, \quad L_X \leq p_X \leq U_X \quad (X \in \mathcal{C})\right\}. \tag{3.9}$$

An extension with profile $\mathbf{p}$ requires a total of $d_X = p_X - k_X$ occurrences of X in the gaps. For a singleton $S = \{X\}$, condition (3.8) is $p_X \leq U_X$. For the two-color set $S = \mathcal{C}\setminus\{X\}$, use $\sum_X d_X = \sum_g \ell_g$ to rewrite the same condition as

$$d_X \geq \sum_g \left(\ell_g - \min\left\{\ell_g, \sum_{Y\neq X} u_{gY}\right\}\right),$$

which is $p_X \geq L_X$. The empty and full color sets impose no additional restrictions. Also $L_X \geq k_X$, so these lower bounds ensure $d_X \geq 0$. Thus (3.8) supplies an integer allocation exactly when (3.9) holds, and Theorem 3.7 realizes each allocated gap by a proper path. This proves the claim.

Both $\mathbf{x}$ and $\mathbf{z}$ satisfy the coordinate intervals in (3.9). Every vector coordinatewise between them satisfies the same intervals. If its total is n, it therefore has the required proper extension. □

In our applications, K will consist of selected agreements with the source. Interpolation keeps these particular agreements fixed. It does not require all agreements of either extension to be retained.

3.3. **Proper source colorings.** Throughout this subsection, w is proper. Hence $1 \le m_X \le r$ and $q_X \ge 1$ for every color. All target profiles satisfy (3.3). The first two target constructions assume $\mathbf{p} \ge \mathbf{q}$; we then treat a deficient coordinate separately.

Call a target color *saturated* if its multiplicity is r, the largest feasible count. A target with r copies of A needs an independent A class of size r. Theorems 3.9 and 3.10 choose that class with enough source A vertices inside it and enough source B, C vertices outside it, then complete the target coloring.

Lemma 3.9. *There is an independent set I of size r such that*

$$|I \cap w^{-1}(A)| \ge q_A,$$
$$|I \cap w^{-1}(B)| \le \left\lfloor \frac{m_B}{2} \right\rfloor, \qquad |I \cap w^{-1}(C)| \le \left\lfloor \frac{m_C}{2} \right\rfloor.$$

Moreover, the unique edge of the subgraph induced by $V(C_n) \setminus I$ has original endpoint colors B and C.

Proof. Index the vertices modulo n, and put

$$I_j = \{j+2, j+4, \dots, j+2r\}.$$

Each I_j is independent, and its complement induces precisely the edge $\{j, j+1\}$. The sets I_j, I_{j+1} are disjoint and together contain every vertex except $j+1$.

Call I_j admissible if its numbers of original B and C vertices satisfy the two asserted upper bounds. An admissible set exists. Otherwise, label each I_j by a color whose bound it violates, choosing either one if both are violated. Two consecutive sets cannot receive the same label: being disjoint, they would then contain at least $2(\lfloor m_X/2 \rfloor + 1) > m_X$ vertices of that color X. The labels would give a proper two-coloring of the odd cycle of indices, a contradiction.

Every admissible set contains at least

$$r - \left\lfloor \frac{m_B}{2} \right\rfloor - \left\lfloor \frac{m_C}{2} \right\rfloor$$

original A vertices. Checking the parities using $m_A + m_B + m_C = 2r + 1$, this number is at least q_A unless m_A is odd and m_B, m_C are even. In that exceptional case it is $q_A - 1$. If an admissible I_j fails the A quota, its three counts must therefore be exactly

$$\left(\frac{m_A - 1}{2}, \frac{m_B}{2}, \frac{m_C}{2} \right).$$

The adjacent sets I_{j-1} and I_{j+1} are disjoint from I_j and omit, respectively, j and $j+1$ from their union with it. At least one of those omitted vertices is not originally A, because w is proper on the edge $\{j, j+1\}$. The corresponding adjacent set contains $(m_A + 1)/2 = q_A$ original A vertices. Its B, C counts are at most $m_B/2, m_C/2$, respectively, so it is admissible. Thus an admissible set meeting the A quota exists in every case.

Choose an admissible I_j maximizing its number of original A vertices. If $w_j = A$, replace I_j by I_{j-2}; this removes $j-1$ and inserts j. Properness gives $w_{j-1} \ne A$, so the A count increases and neither of the other counts increases. This contradicts maximality. Similarly, if $w_{j+1} = A$, replacing I_j by I_{j+2} removes $j+2$ and inserts $j+1$, with the same contradiction. Thus neither endpoint of the complementary edge is originally A. Their original colors are distinct, so they are B, C. □

Lemma 3.10. *Suppose $\mathbf{p} \ge \mathbf{q}$. If a target multiplicity equals r, a proper source has a proper target satisfying all agreement quotas.*

Proof. Suppose $p_A = r$, and color every vertex of the set I in Theorem 3.9 with A. This retains at least q_A original A vertices. The complement of I contains at least $m_X - \lfloor m_X/2 \rfloor = q_X$ original vertices of color X, for each $X \in \{B, C\}$.

In that complement, select q_B original B vertices and q_C original C vertices to retain. Include the corresponding endpoints of its unique edge; this is possible because $q_B, q_C \ge 1$. There remain $p_B - q_B$ additional B assignments and $p_C - q_C$ additional C assignments. These numbers are nonnegative and their sum equals the number of unassigned vertices, so fill those vertices accordingly. The sole edge internal to the complement is proper, and every edge from the complement to I joins B or C to A. □

For a smaller increase in one target color, we use a saturated target as an endpoint and interpolate back toward the source profile.

Lemma 3.11. *Suppose $\mathbf{p} \ge \mathbf{q}$ and $p_X > m_X$ for at most one color X. A proper source then has a proper target satisfying all agreement quotas.*

Proof. If no coordinate increases, the equality of the totals forces $\mathbf{p} = \mathbf{m}$, and w itself suffices. Otherwise suppose A is the sole increasing color. We construct a profile $\mathbf{P} = (r, P_B, P_C)$ with

$$q_B \le P_B \le p_B, \qquad q_C \le P_C \le p_C, \qquad P_B + P_C = r + 1.$$

Choose an integer P_B in the interval

$$\max\{q_B, r + 1 - p_C\} \le P_B \le \min\{p_B, r + 1 - q_C\}, \qquad P_C = r + 1 - P_B.$$

This interval is nonempty. Indeed, $q_B \le p_B$ and $q_C \le p_C$ hold by assumption, while

$$p_B + p_C = 2r + 1 - p_A \ge r + 1, \qquad q_B + q_C \le r + 1.$$

The last inequality also follows from (3.5). These four inequalities compare each lower endpoint with each upper endpoint. The resulting $\mathbf{P}$ is feasible and dominates all quotas: its B, C entries lie below the corresponding entries of $\mathbf{p}$, and $q_A \le m_A \le r = P_A$.

Theorem 3.10 gives a proper coloring z of profile $\mathbf{P}$ meeting the quotas. Select exactly q_X agreements between w and z of each color X, and let K be their union, with the common colors prescribed there. Both w and z are proper extensions of K. Furthermore,

$$m_A < p_A \le P_A, \qquad P_B \le p_B \le m_B, \qquad P_C \le p_C \le m_C.$$

Thus $\mathbf{p}$ lies coordinatewise between $\mathbf{m}$ and $\mathbf{P}$. Theorem 3.8 supplies a proper extension of K with profile $\mathbf{p}$. Its agreements on K meet every quota. □

We also need a direct construction when the target has too few copies of one color to meet its full half-count requirement.

Lemma 3.12. *If $p_A \le q_A$, a proper source has a proper target satisfying all agreement quotas.*

Proof. By (3.5), $p_B \ge q_B$ and $p_C \ge q_C$. The required quotas are therefore $t_A = p_A$, $t_B = q_B$, and $t_C = q_C$. Set $a = m_A$ and $h = p_A - 1$. Apply Theorem 3.5 to the original A class. Keep the chosen root s colored A. The two parity classes of $C_n - s$ each have r vertices, and their original A counts differ by at most one. Each therefore contains at least $\lfloor (a-1)/2 \rfloor = q_A - 1 \ge h$ original A positions.

Temporarily color the path alternately B, C. To choose between the two alternating colorings, ignore the original A positions when counting agreements. The two sets of remaining original B, C vertices have sizes differing by at most one, because the full parity classes have equal size and their original A counts differ by at most one. Theorem 3.4 therefore gives a choice retaining at least q_B original B vertices and q_C original C vertices. Together with the root, this is a proper coloring of profile $(1, r, r)$.

Now recolor to A precisely $r - p_B$ original A positions currently colored B, and $r - p_C$ original A positions currently colored C. The required numbers are nonnegative, and

$$(r - p_B) + (r - p_C) = 2r - (2r + 1 - p_A) = p_A - 1 = h.$$

Each number is therefore at most h, and each parity class contains enough original A positions. Including the root, all newly A-colored vertices lie in the original A class, which is independent. Thus the recoloring remains proper. Its profile is $\mathbf{p}$, and it retains exactly $p_A = t_A$ of the original A vertices. None of the selected B, C agreements has been changed. □

The preceding lemmas leave only targets that increase two color counts from a proper source. The boundary case $P_A = q_A$ of Theorem 3.12 provides the second endpoint for interpolation.

Proposition 3.13. *Theorem 3.3 holds whenever the source is proper.*

Proof. If some coordinate is deficient, relabel it A and apply Theorem 3.12. We may therefore assume $\mathbf{p} \ge \mathbf{q}$, so the agreement quotas are the half-count requirements q_X. Theorem 3.11 treats every target with at most one increasing coordinate. In the remaining case, precisely two coordinates increase and the third decreases. Relabel so that

$$p_B > m_B, \qquad p_C > m_C, \qquad p_A < m_A.$$

We first construct a target on the boundary $P_A = q_A$, further in the same coordinate directions than $\mathbf{p}$. Set $t = p_A - q_A \ge 0$. Distribute these t units between the two increasing colors by choosing an integer u satisfying

$$0 \le u \le r - p_B, \qquad 0 \le t - u \le r - p_C.$$

Such a choice exists because the total available capacity is

$$(r - p_B) + (r - p_C) = p_A - 1 \ge p_A - q_A = t.$$

For example, one may take any integer in $[\max\{0, t-(r-p_C)\}, \min\{t, r-p_B\}]$.

The vector

$$\mathbf{P} = (q_A, p_B + u, p_C + t - u)$$

has total $2r+1$, every coordinate lies between 1 and r, and every coordinate dominates the corresponding quota. Theorem 3.12 gives a proper coloring z of this profile meeting all quotas. As before, choose exactly q_X agreements of each color between w and z and prescribe their common colors on the resulting nonempty set K.

Both w and z are proper extensions of K. The inequalities

$$P_A = q_A \le p_A < m_A, \qquad m_B < p_B \le P_B, \qquad m_C < p_C \le P_C$$

place $\mathbf{p}$ coordinatewise between their profiles. Theorem 3.8 gives a proper extension with profile $\mathbf{p}$, and the fixed agreements on K supply the quotas. □

3.4. **Induction for arbitrary sources.** For an improper source, an equal adjacent pair supplies the two vertices deleted in the induction. The target counts are reduced in two colors so that reinserting the pair restores both its profile and its agreement quotas.

Proof of Theorem 3.3. We induct on the odd cycle length $n = 2r+1$. For $n = 3$, the target profile must be $(1,1,1)$ and every source color that occurs has quota one. Retain one vertex of each such color, then assign the missing colors to the remaining vertices. The resulting coloring uses all three colors once, hence is proper and meets every quota, including for a monochromatic source.

Suppose $n \ge 5$. Theorem 3.6 handles targets having a coordinate equal to one, and Theorem 3.13 handles proper sources. We may therefore assume that the source contains an equal adjacent pair and that

$$2 \le p_X \le r \quad (X \in \mathcal{C}).$$

At most one target count equals r, since two such counts would force the third to be one. In particular, the remaining case has $r \ge 3$.

We first show that, for every source color A occurring on an equal adjacent pair, some different color B has

$$p_B > q_B. \tag{3.10}$$

Indeed, the repeated pair gives $m_A \ge 2$, so the other two source counts have total at most $2r-1$. Consequently

$$q_B + q_C \le \left\lceil \frac{m_B + m_C + 1}{2} \right\rceil \le r, \qquad p_B + p_C = 2r + 1 - p_A \ge r + 1.$$

At least one of those two target counts strictly exceeds its half-count requirement; label that color B. This proves (3.10).

Choose an equal adjacent pair AA and a different slack color B so that any target color of multiplicity r is one of A, B. If no target multiplicity is r, there is nothing further to arrange. If the unique such color is X and the source contains an adjacent XX, take $A = X$ and use (3.10) to choose B. If the source contains no XX, its X vertices form an independent set, so

$$m_X \le r, \qquad q_X \le \left\lceil \frac{r}{2} \right\rceil < r = p_X.$$

Choose any equal adjacent pair AA and take $B = X$. This also has the required properties.

Delete the two vertices of the selected pair, and join their outside neighbors to obtain C_{n-2}. Its source profile is

$$\mathbf{m}' = \mathbf{m} - 2\mathbf{e}_A.$$

Prescribe the smaller target profile

$$\mathbf{p}' = \mathbf{p} - \mathbf{e}_A - \mathbf{e}_B,$$

where $\mathbf{e}_X$ is the coordinate unit vector for X. The new half-count requirements satisfy

$$\mathbf{q}' = \mathbf{q} - \mathbf{e}_A,$$

because $\lceil (m_A - 2)/2 \rceil = \lceil m_A/2 \rceil - 1$. For coordinate A, the target count and half-count requirement both decrease by one. Thus its capped quota also decreases by one:

$$t'_A = \min\{p_A - 1, q_A - 1\} = t_A - 1.$$

For coordinate B, the strict integer inequality (3.10) gives $p_B - 1 \geq q_B$, so $t'_B = q_B = t_B$. The third capped quota is unchanged. Hence $\mathbf{t}' = \mathbf{t} - \mathbf{e}_A$. Every target count remains positive. Every coordinate originally equal to r has been reduced, and all other coordinates were already at most $r-1$. Thus

$$\sum_X p'_X = 2(r-1)+1, \qquad 1 \leq p'_X \leq r-1.$$

All induction hypotheses hold. Zero coordinates of the reduced source cause no problem because the theorem allows arbitrary nonnegative source counts.

Apply induction to obtain a proper target coloring y' on the smaller cycle meeting the reduced quotas. Let L, R be its colors at the endpoints of the new edge. Since y' is proper, $L \neq R$. Replace that edge by the original two vertices, assigning them either AB or BA. The order AB fails only if $L = A$ or $R = B$; the order BA fails only if $L = B$ or $R = A$. If both orders failed, their two conditions, together with $A \neq B$, would force either $L = R = A$ or $L = R = B$. This is impossible, so at least one order produces a proper coloring.

The insertion restores the profile $\mathbf{p}$. Both restored source vertices were originally A, and exactly one has now been colored A. We have therefore added one A agreement, restoring its quota from $t_A - 1$ to t_A. All previously retained vertices keep their colors, so no other quota is lost. This completes the induction. □

3.5. **Directed profile radii.**

Definition 3.14 (Directed profile radius)**.** For any nonnegative integer source profile $\mathbf{m}$ of total n and any feasible target profile $\mathbf{p}$, put

$$R_{C_n}(\mathbf{m} \to \mathbf{p}) = \max_{\substack{w: V(C_n) \to \mathcal{C} \\ \mathbf{m}(w) = \mathbf{m}}} \min_{y \in \mathcal{P}_{\mathbf{p}}(C_n)} d_{\mathrm{H}}(w, y).$$

The source profile need not be feasible. For a nonempty family $\mathcal{Q}$ of feasible target profiles, define $R_{C_n}(\mathbf{m} \to \mathcal{Q})$ by allowing any proper target whose profile belongs to $\mathcal{Q}$ in the same maximum–minimum.

Write $(x)_+ = \max\{x, 0\}$.

Theorem 3.15 (Directed and restricted-target radii)**.** *For every pair in Theorem 3.14, with $n = 2r+1$,*

$$\begin{aligned} R_{C_{2r+1}}(\mathbf{m} \to \mathbf{p}) &= 2r + 1 - \sum_{X \in \mathcal{C}} \min\left\{p_X, \left\lceil \frac{m_X}{2} \right\rceil\right\} \\ &= \sum_{X \in \mathcal{C}} \left\lfloor \frac{m_X}{2} \right\rfloor + \sum_{X \in \mathcal{C}} \left(\left\lceil \frac{m_X}{2} \right\rceil - p_X \right)_+ . \end{aligned}$$

At most one term in the positive-part sum is nonzero. More generally, for every nonempty family $\mathcal{Q}$ of feasible target profiles,

$$R_{C_{2r+1}}(\mathbf{m} \to \mathcal{Q}) = 2r + 1 - \max_{\mathbf{p} \in \mathcal{Q}} \sum_{X \in \mathcal{C}} \min\{p_X, \lceil m_X/2 \rceil\}. \tag{3.11}$$

The block word $A^{m_A} B^{m_B} C^{m_C}$ attains the radius for every target profile simultaneously. In particular, every feasible source profile satisfies

$$R^{\mathrm{w}}_{C_{2r+1}}(\mathbf{m}) = R_{C_{2r+1}}(\mathbf{m}) = \sum_{X \in \mathcal{C}} \left\lfloor \frac{m_X}{2} \right\rfloor .$$

Proof. The upper bound is Theorem 3.3. For the reverse inequality, use the three-block obstruction of Clark et al. [5, Proposition 3.5], allowing empty blocks. If the source is not monochromatic, each nonempty block induces a path, so a proper target retains at most $\lceil m_X/2 \rceil$ vertices of color X. Its target profile also limits that number to p_X. Summing gives the lower bound. For a monochromatic source, every target has exactly p_X agreements in its sole source color, and the same formula follows directly. The same block word gives these lower bounds simultaneously for all $\mathbf{p} \in \mathcal{Q}$. For the reverse bound, apply Theorem 3.3 with a profile in $\mathcal{Q}$ maximizing the displayed agreement sum. This proves (3.11). The second expression is an algebraic rearrangement, and (3.5) proves the assertion about the positive-part sum. For a feasible source profile, take $\mathbf{p} = \mathbf{m}$ to obtain the ordinary value. The same block obstruction bounds the weak radius from below, while the ordinary upper bound applies to the larger family of weak targets. Hence the two profile radii agree. □

4. Complete multipartite graphs

On an odd cycle, Theorem 3.15 determines the repair cost for each pair of source and target profiles. For a complete multipartite graph, every optimal proper coloring is constant on each part, so repair instead amounts to assigning colors to parts. Prescribed multiplicities become row and column sums of a disturbance matrix. The transportation criterion in Theorem 2.3, which governed the gap allocations in Theorem 3.8, will again determine which integer counts can be realized. The resulting matrix model also provides the starting point for the extrema in Section 5.

4.1. **Matrix representation and assignment duality.** For a nonnegative square matrix $M = (m_{ij})$ of order h, define its *assignment score*, or max-plus tropical determinant, by

$$\tau(M) = \max_{\pi \in S_h} \sum_{i=1}^{h} m_{i,\pi(i)}.$$

Let $G = K_{p_1,\dots,p_k} = K_{\mathbf{p}}$, where every p_i is positive, and put $\mathbf{p} = (p_1, \dots, p_k)^{\mathsf{T}}$ and $n = \sum_i p_i$. Every proper k-coloring of G assigns a distinct color to each part. We may therefore label the colors so that color j has multiplicity p_j. For a rearrangement w, let m_{ij} count the vertices of color j in part i. The possible matrices are precisely

$$\mathcal{T}(\mathbf{p}) = \{M \in \mathbb{Z}_{\geq 0}^{k \times k} : M\mathbf{1} = \mathbf{p},\ M^{\mathsf{T}}\mathbf{1} = \mathbf{p}\}.$$

We call M the *disturbance matrix* of w; its row and column sum vectors are its *margins*. Recall that $\tau(M)$ is the maximum assignment score. Assigning color $\pi(i)$ to part i retains exactly $\sum_i m_{i,\pi(i)}$ vertices. Consequently,

$$B_{\mathrm{w}}(G) = n - \mu(\mathbf{p}), \qquad \mu(\mathbf{p}) = \min_{M \in \mathcal{T}(\mathbf{p})} \tau(M). \tag{4.1}$$

For a repair preserving color multiplicities, the assignment is allowed only when $p_i = p_{\pi(i)}$ for every i. Partition $[k]$ into *size classes*, grouping indices with equal part sizes. If $M[\Gamma, \Gamma]$ is the principal submatrix belonging to a size class Γ, the allowed assignments act independently on these classes. Thus

$$B(G) = n - \min_{M \in \mathcal{T}(\mathbf{p})} \sum_{\Gamma} \tau(M[\Gamma, \Gamma]). \tag{4.2}$$

The distinction between the two parameters is therefore a distinction between unrestricted assignments and assignments constrained by the size classes.

Row and column *potentials* are vectors $\mathbf{u}, \mathbf{v}$. They *cover* M if $u_i + v_j \geq m_{ij}$ for every i, j, and their *cost* is $\sum_i u_i + \sum_j v_j$. We shall use the following integral form of assignment duality [14]. We include a short proof to fix the normalization of its nonnegative potentials.

Lemma 4.1 (Integral assignment duality)**.** *For a nonnegative integer matrix M of order h,*

$$\tau(M) = \min \left\{ \sum_i u_i + \sum_j v_j : \mathbf{u}, \mathbf{v} \in \mathbb{Z}_{\geq 0}^{h}, \quad u_i + v_j \geq m_{ij} \text{ for every } i, j \right\}.$$

Proof. Every feasible pair of potentials bounds every assignment score by its total sum. To obtain equality, choose an optimal permutation and permute the columns to make its entries diagonal; call the resulting matrix H. Give each directed edge $i \to j$ weight $h_{ij} - h_{ii}$. A directed cycle of positive weight would replace part of the diagonal assignment by a larger assignment. Thus every directed cycle has nonpositive weight.

Adjoin a source with a zero-weight edge to every index and let a_j be the maximum weight of a walk from this source to j. Removing a closed subwalk cannot decrease its weight, so the maximum is attained on a simple path. In particular, every a_j is a finite integer, and

$$a_j \geq a_i + h_{ij} - h_{ii}.$$

The integer potentials $u_i = h_{ii} - a_i$ and $v_j = a_j$ cover H and have total cost $\sum_i h_{ii} = \tau(M)$. Restore the original column order to obtain potentials covering M with the same cost.

These potentials may be signed. Put $c = \min_i u_i$ and replace them by $u_i - c$ and $v_j + c$. The first vector is nonnegative. A row attaining $u_i = c$ gives $v_j + c \geq m_{ij} \geq 0$ for every j, so the second vector is also nonnegative. The covering inequalities and total cost are unchanged. □

4.2. **Ordinary villainy and the dominant size class.** A size class consisting of m parts of size d contains md vertices. It is *dominant* when $md > n/2$; at most one such class exists. The next theorem evaluates the contribution of this class. The size-class reduction and its maximum-weight matching interpretation already appear in Raleigh's thesis [16, Chapter II].

We first identify a known extremal case. Let $\mathfrak{G}$ be the subgroup of permutations of $V(G)$ that map each part bijectively onto a part of the same size. Its Hamming covering radius in the full symmetric group is the largest distance of a vertex permutation from its nearest element of $\mathfrak{G}$. This radius equals $B(G)$. Indeed, fix a proper coloring c constant on the parts. Every rearrangement has the form $c \circ \pi$, and every count-preserving proper target has the form $c \circ \gamma$ with $\gamma \in \mathfrak{G}$. For a fixed target, retain the partial bijection $\gamma(v) = \pi(v)$ at every color agreement and extend it within the corresponding parts. This extension has exactly those point agreements. Thus the minimum permutation distance equals the minimum color distance, and the maxima also agree. The orbits of $\mathfrak{G}$ are exactly the unions of all parts of each fixed size. Cameron and Wanless [4, Theorem 14] therefore give the following criterion. Raleigh [16, Theorem 1] also proved the full-change conclusion directly when no size class contains more than half the vertices.

Proposition 4.2 (Cameron–Wanless)**.** *For a complete multipartite graph G on n vertices, $B(G) = n$ if and only if every size class contains at most $n/2$ vertices.*

Theorem 4.3. *If every size class contains at most $n/2$ vertices, then $B(G) = n$. Otherwise, let (m, d) be the parameters of the unique size class with $md > n/2$, and put*

$$P = n - md, \qquad t = md - P, \qquad b = d - m\left\lfloor \frac{d}{m} \right\rfloor.$$

If $P \geq mb$, then

$$B(G) = n - \left\lceil \frac{t}{m} \right\rceil \qquad \text{[16, Theorem 4].} \tag{4.3}$$

For the remaining case $P < mb$, and for $0 \leq r, s < m$, set $h = m - r$, $\ell = m - s$, and

$$\begin{aligned} q_{r,s} = \max\Bigg\{0, \left\lceil \frac{t - m(r+s)}{m^2} \right\rceil, \left\lceil \frac{t - dr - hs}{mh} \right\rceil, \\ \left\lceil \frac{t - ds - \ell r}{m\ell} \right\rceil, \left\lceil \frac{t - d(r+s)}{h\ell} \right\rceil \Bigg\}. \end{aligned} \tag{4.4}$$

Then

$$B(G) = n - F(m, d, t), \qquad F(m, d, t) = \min_{0 \leq r, s < m} (mq_{r,s} + r + s). \tag{4.5}$$

In particular, the formula uses at most k^2 arithmetic candidates.

Raleigh [16, Theorems 2–3] obtained bounds in the remaining regime $P < mb$. Formula (4.5) evaluates that regime. The uniform-part problem was determined independently by Dinitz, Hartman, and Soprunova [6]; its extension to rectangular transportation matrices is given in [9]. We do not give separate proofs of these previously known formulas.

We isolate the matrix problem behind the remaining case. For integers $0 \leq t \leq md$, define

$$F(m, d, t) = \min\left\{ \tau(H) : \begin{array}{l} H \in \mathbb{Z}_{\geq 0}^{m \times m}, \quad H\mathbf{1} \leq d\mathbf{1}, \quad H^{\mathsf{T}}\mathbf{1} \leq d\mathbf{1}, \\ \displaystyle\sum_{i,j} h_{ij} = t \end{array} \right\}. \tag{4.6}$$

The two meanings of F in (4.5) and (4.6) will be shown to agree.

Lemma 4.4. *If a size class with parameters (m, d) has $md > n/2$, then $B(G) = n - F(m, d, 2md - n)$, with F defined by (4.6).*

Proof. Write $S = md$, $P = n - S$, and $t = S - P$. In any matrix $M \in \mathcal{T}(\mathbf{p})$, the block H belonging to this class has total sum at least $S - P = t$: its rows have total S, whereas the columns outside the class can receive only P units. Its row and column sums are at most d. Delete units from its entries until its total is t. This operation cannot increase an assignment score, so $\tau(H) \geq F(m, d, t)$. The other diagonal blocks contribute nonnegative scores to (4.2).

Conversely, take a matrix H in (4.6). Its row deficits from d sum to $md - t = P$, exactly the total demand of the outside columns. Fill the block from the internal rows to the outside columns with these margins. Similarly, fill the block from the outside rows to the internal columns using the

column deficits of H. Both fillings exist with integer entries: one can repeatedly place the smaller of a remaining row supply and column demand in their intersection. Set the outside-to-outside block to zero. The resulting matrix belongs to $\mathcal{T}(\mathbf{p})$, and H is its only nonzero diagonal size-class block. Minimizing over H proves the assertion. □

The following elementary balancing fact explains why only two values are needed in each vector of assignment potentials.

Lemma 4.5. *Among nonnegative integer vectors of a fixed length and sum, a vector whose entries differ by at most one maximizes the sum of the smallest j entries, simultaneously for every j.*

Proof. Suppose a largest entry a and a smallest entry b satisfy $a \geq b+2$. Transfer one unit from a to b. Among the subsets of j entries with minimum sum after the transfer, choose one that does not contain the donor without the receiver. Such a choice is possible because the receiver's new value $b+1$ is at most the donor's new value $a-1$. On this chosen subset the transfer has not decreased the sum. Its old sum was at least the old minimum, so the minimum sum of j entries has not decreased. This holds for every j. Repeating the operation terminates, since the sum of squares decreases at each step, and the final vector has entries differing by at most one. □

Proof of Theorem 4.3. The case without a dominant size class is Theorem 4.2, and the case $P \geq mb$ is Raleigh's theorem quoted in (4.3).

It remains to treat $md > n/2$ and $P < mb$. By Theorem 4.4, it remains to evaluate (4.6). For nonnegative integer vectors $\mathbf{u}, \mathbf{v}$ of length m, let $\Phi(\mathbf{u}, \mathbf{v})$ be the largest total sum of an integer matrix whose row and column sums are at most d and whose (i,j) entry is at most $u_i + v_j$. Integral assignment duality, Theorem 4.1, gives

$$F(m,d,t) = \min\left\{ \sum_i u_i + \sum_j v_j : \mathbf{u}, \mathbf{v} \in \mathbb{Z}^m_{\geq 0},\ \Phi(\mathbf{u}, \mathbf{v}) \geq t \right\}. \tag{4.7}$$

Indeed, optimal potentials for a matrix in (4.6) give a feasible pair on the right with the same assignment value. Conversely, if the capacities admit at least t units, an integral flow can be reduced to exactly t units. The resulting matrix has every assignment score bounded by the total potential cost.

In the network defining Φ, source-to-row and column-to-sink capacities are all d. A cut with source-side rows I and sink-side columns J has capacity

$$d(2m - |I| - |J|) + |J| \sum_{i \in I} u_i + |I| \sum_{j \in J} v_j. \tag{4.8}$$

For fixed $|I|$ and $|J|$, the minimum is obtained by choosing the smallest potentials on both sides. Theorem 4.5 therefore shows that balancing $\mathbf{u}$ and $\mathbf{v}$ separately cannot decrease any minimum cut and hence cannot decrease Φ. It preserves the objective in (4.7).

Write the balanced row potentials as q_1 and q_1+1, with r entries of the larger value, and the balanced column potentials as q_2 and q_2+1, with s entries of the larger value. Here $0 \leq r, s < m$. Put

$$q = q_1 + q_2, \qquad h = m - r, \qquad \ell = m - s, \qquad T = mq + r + s.$$

The entry capacities are $q, q+1, q+2$, and their total potential cost is T. Every nonnegative integer q is possible, for example by choosing $q_1 = q$ and $q_2 = 0$.

The sum of the smallest i row potentials is $iq_1 + \max(0, i-h)$; the corresponding column sum is $jq_2 + \max(0, j-\ell)$. Substituting these expressions into (4.8) gives a bilinear function of i, j on each rectangle separated by the lines $i = h$ and $j = \ell$. On a rectangle, a bilinear function has a minimum at a corner: first minimize in one coordinate and then in the other. It is thus enough to test $i \in \{0, h, m\}$ and $j \in \{0, \ell, m\}$. The resulting flow value is

$$\begin{aligned} \Phi(q,r,s) = \min\{&md,\ mT,\ dr + (m-r)(T-r), \\ &ds + (m-s)(T-s),\ d(r+s) + q(m-r)(m-s)\}. \end{aligned} \tag{4.9}$$

Because $t \leq md$, the first expression imposes no restriction. The other four inequalities $\Phi(q,r,s) \geq t$ give, after solving for the integer $q \geq 0$, precisely (4.4). Substitution into (4.7) proves (4.5). □

For distinct part sizes, the theorem reduces to the formula

$$B(K_{p_1,\dots,p_k}) = \min\{n, 2(n - \max_i p_i)\},$$

proved by Clark et al. [5, Proposition 4.1]. We also record the complete bipartite specialization for later use; it is a special case of their connected bipartite formula.

Proposition 4.6 (Clark et al. [5])**.** *If $1 \le a < b$, then $B(K_{a,b}) = 2a$. For $a \ge 1$, $B(K_{a,a}) = 2\lfloor a/2 \rfloor$.*

4.3. **Weak villainy and ordered potentials.** For weak repairs, every permutation is allowed, and the separate size classes no longer decouple. Nevertheless, the transportation formulation gives a finite formula with substantially fewer variables than the original matrix problem.

Assume $p_1 \le \cdots \le p_k$, and define

$$\mathcal{H}(\mathbf{p}) = \left\{ \mathbf{h} \in \mathbb{Z}^k_{\ge 0} : \begin{array}{l} 0 = h_1 \le \cdots \le h_k, \\ h_j - h_i \le \max\{1, p_j - p_i\} \quad (1 \le i < j \le k) \end{array} \right\}. \tag{4.10}$$

We call the elements of $\mathcal{H}(\mathbf{p})$ *potential shapes*. In particular, $h_i \le p_i$, and entries belonging to one size class differ by at most one. For a real vector $\mathbf{h}$ and $1 \le r, s \le k$, let $A_{r,s}(\mathbf{h})$ be the sum of the largest r entries of $(p_i - sh_i)_{i=1}^k$. In particular, negative entries are retained in this definition. For $\mathbf{h}, \boldsymbol{\ell} \in \mathcal{H}(\mathbf{p})$, put

$$Q(\mathbf{h}, \boldsymbol{\ell}) = \max\left\{ 0, \max_{1 \le r,s \le k} \left\lceil \frac{A_{r,s}(\mathbf{h}) + A_{s,r}(\boldsymbol{\ell}) - n}{rs} \right\rceil \right\}. \tag{4.11}$$

Theorem 4.7. *Let $\mathbf{p} = (p_1, \dots, p_k)$ with $1 \le p_1 \le \cdots \le p_k$ and $n = \sum_i p_i$. Then $B_{\mathrm{w}}(K_{\mathbf{p}}) = n - \mu(\mathbf{p})$, where*

$$\mu(\mathbf{p}) = \min_{\mathbf{h}, \boldsymbol{\ell} \in \mathcal{H}(\mathbf{p})} \left(kQ(\mathbf{h}, \boldsymbol{\ell}) + \sum_i h_i + \sum_i \ell_i \right). \tag{4.12}$$

An attaining pair of vectors determines an extremal rearrangement by an integral flow construction.

The required reduction of the potentials uses the following exchange argument.

Lemma 4.8. *Suppose $0 \le u_i, v_i \le p_i$ are integers and some matrix in $\mathcal{T}(\mathbf{p})$ satisfies $m_{ij} \le u_i + v_j$. Without changing the sum of either vector or losing the existence of such a matrix, the potentials can be chosen so that, for every $i < j$,*

$$0 \le u_j - u_i \le \max\{1, p_j - p_i\}, \qquad 0 \le v_j - v_i \le \max\{1, p_j - p_i\}.$$

The bounds $0 \le u_i, v_i \le p_i$ are preserved.

Proof. By Theorem 2.3, the capacities $u_i + v_j$ admit a matrix in $\mathcal{T}(\mathbf{p})$ exactly when

$$p(I) + p(J) - n \le |J| u(I) + |I| v(J) \qquad (I, J \subseteq [k]), \tag{4.13}$$

where $p(I) = \sum_{i \in I} p_i$, and similarly for $u(I), v(I)$. For $|I| = r > 0$ and $|J| = s > 0$, the largest difference between the two sides is $A_{r,s}(\mathbf{u}) + A_{s,r}(\mathbf{v}) - n$. The choices of I and J are independent. Conditions with an empty set hold automatically.

Suppose $i < j$ but $u_i > u_j$, and interchange u_i, u_j. For any positive integer s, the old residuals

$$a = p_i - su_i, \qquad b = p_j - su_j$$

satisfy $a \le b$. The two new residuals have the same sum and both lie in $[a, b]$. Replacing a pair by two numbers of the same sum in their old interval cannot increase the sum of the largest r entries of the whole vector. Indeed, a maximizing r-subset uses zero, one, or both entries. Only the middle case needs consideration, and its contribution is at most the larger old entry. No $A_{r,s}(\mathbf{u})$ increases, so (4.13) remains valid. The coordinate bounds also remain valid: the potential moved to i decreases, and the potential moved to j is at most $p_i \le p_j$. Repeating removes all inversions.

After sorting, suppose that $i < j$ and $u_j - u_i > \max\{1, p_j - p_i\}$. Put $\delta = p_j - p_i$ and $\eta = u_j - u_i$, and transfer one unit from u_j to u_i. For every positive integer s, the old residuals $a = p_i - su_i$ and $b = p_j - su_j$ satisfy

$$a - b = s\eta - \delta \ge s,$$

because $\eta \ge \delta + 1$. The new residuals are $a - s$ and $b + s$; both lie in $[b, a]$ and their sum is unchanged. The same contraction argument therefore preserves every cut inequality. The coordinate bounds survive as well: $u_i = p_i$ would force $u_j > p_j$, and $u_j \ge u_i + 2$ ensures that the donor stays nonnegative.

Sort the modified vector again, and repeat whenever a pair violates the asserted bound. Each transfer strictly decreases the sum of squares, whereas sorting leaves it unchanged, so the procedure terminates. The sum of the vector and feasibility are preserved at every step. Apply the procedure first to $\mathbf{u}$ and then to $\mathbf{v}$. □

Proof of Theorem 4.7. Choose $M \in \mathcal{T}(\mathbf{p})$ attaining $\mu(\mathbf{p})$ and nonnegative integral optimal assignment potentials $\mathbf{u}, \mathbf{v}$. We may assume $u_i \le p_i$ and $v_j \le p_j$. To see this, truncate each potential at its corresponding margin. If u_i is truncated, its new value p_i alone is at least every entry in row i; the other potential is nonnegative. The same argument applies to a truncated column potential. Thus truncation preserves the covering inequalities and does not increase the cost.

Apply Theorem 4.8. The modified potentials need not cover the original matrix, but they cover some matrix with the same margins, which is sufficient for minimizing $\mu(\mathbf{p})$. Subtract the minimum separately from each vector:

$$\alpha = \min_i u_i, \quad \beta = \min_j v_j, \quad q = \alpha + \beta, \qquad h_i = u_i - \alpha, \quad \ell_j = v_j - \beta.$$

Subtracting a constant preserves all difference bounds, so $\mathbf{h}, \boldsymbol{\ell} \in \mathcal{H}(\mathbf{p})$. The capacities become $q+h_i+\ell_j$, and the cost is $kq + \sum_i h_i + \sum_j \ell_j$.

For fixed shapes $\mathbf{h}, \boldsymbol{\ell}$, substitute these capacities into (4.13). For nonempty I, J of cardinalities r, s, the condition becomes

$$rsq \ge p(I) - sh(I) + p(J) - r\ell(J) - n.$$

Maximizing the right side independently over I and J shows that all cuts are feasible exactly when $q \ge Q(\mathbf{h}, \boldsymbol{\ell})$. Decreasing q to this least value therefore preserves feasibility. This proves that the right side of (4.12) is at most $\mu(\mathbf{p})$.

Conversely, any shapes with $q = Q(\mathbf{h}, \boldsymbol{\ell})$ satisfy all capacity inequalities. Theorem 2.3 gives an integer matrix in $\mathcal{T}(\mathbf{p})$ bounded by these capacities. Every permutation score is at most $kq + \sum_i h_i + \sum_j \ell_j$. Hence $\mu(\mathbf{p})$ is at most each candidate in (4.12), proving equality. For a minimizing pair, the constructed matrix has assignment value exactly $\mu(\mathbf{p})$ and therefore yields the asserted extremal rearrangement. □

The difference bounds also control the number of candidates in terms of the spread of the part sizes. A separate consequence of the transportation formulation is fixed-parameter tractability in the number of parts, without a bound on that spread.

Corollary 4.9. *Put* $\Delta = \max\{1, p_k - p_1\}$. *The formula in* (4.12) *uses at most*

$$\binom{k + \Delta - 1}{\Delta}^2$$

candidate pairs. In particular, at most k^2 *pairs suffice when the part sizes differ by at most one. Moreover, both* $B(K_{\mathbf{p}})$ *and* $B_{\mathrm{w}}(K_{\mathbf{p}})$ *can be computed in time* $f(k)L^c$, *where* L *is the binary encoding length of* $\mathbf{p}$, f *is computable, and* c *is an absolute constant. An extremal disturbance matrix for weak villainy can be constructed within such a bound.*

Proof. Every shape has first entry zero and all entries at most Δ. There are $\binom{k+\Delta-1}{\Delta}$ nondecreasing choices for its remaining $k-1$ entries in $\{0, \dots, \Delta\}$. The defining difference bounds can only reduce this number. If $p_k - p_1 \le 1$, then $\Delta = 1$.

For the algorithmic assertion, ordinary villainy is given by Theorem 4.3. For weak villainy, minimize $\sum_i u_i + \sum_j v_j$ over integer potentials satisfying

$$0 \le u_i, v_i \le p_i \quad (1 \le i \le k)$$

and all the inequalities (4.13). The optimum of this integer program is $\mu(\mathbf{p})$. Indeed, an optimal disturbance matrix has bounded integral assignment potentials, by the truncation argument in the preceding proof. Conversely, the transportation criterion produces a matrix with these margins whose assignment score is at most the potential cost.

The program has $2k$ variables and at most $4^k + 4k$ inequalities. Its encoding length is bounded by a function of k times a polynomial in L. Fixed-dimensional integer programming [15, 11], with binary search on the integer objective in $[0, n]$, therefore gives the asserted running time. From optimal potentials, a polynomial-time integral maximum-flow algorithm constructs an attaining matrix. This output records the disturbance by its k^2 entries; expanding it into a color on every vertex is unnecessary. □

If all part sizes are equal, the matrix problem is the uniform-margin problem solved in [6, 9]. The present formulation permits arbitrary positive margins.

5. Extremal villainy at fixed order and chromatic number

For integers $1 \le k \le n$, define

$$\mathcal{M}(n,k) = \max\{B(G) : |V(G)| = n,\ \chi(G) = k\},$$
$$\mathcal{M}_{\mathrm{w}}(n,k) = \max\{B_{\mathrm{w}}(G) : |V(G)| = n,\ \chi(G) = k\}.$$

We now use the multipartite description in Section 4 to determine both maxima over all graphs of prescribed order and chromatic number. Theorem 5.1 reduces the problem to complete multipartite graphs. The ordinary extremum then uses the size-class criterion in Theorem 4.2; the weak extremum uses the assignment model (4.1) and the integral duality of Theorem 4.1. In the weak problem, balance of the disturbance matrix imposes a further parity restriction.

The completion must preserve a proper coloring from which the disturbance arose: arbitrary edge addition need not preserve the set of legal disturbances.

Lemma 5.1 (Completion relative to a source coloring)**.** *For $F \in \{B, B_{\mathrm{w}}\}$,*

$$\max_{\substack{|V(G)|=n\\ \chi(G)=k}} F(G) = \max_{\substack{p_1+\cdots+p_k=n\\ p_i\ge 1}} F(K_{p_1,\dots,p_k}).$$

Proof. Fix a graph G with n vertices and chromatic number k. Choose a legal disturbance w attaining $F(G)$, and choose a proper k-coloring f whose labeled color counts agree with those of w. Write $V_1, \dots, V_k$ for the color classes of f. They are nonempty, because $\chi(G) = k$.

Add all missing edges between distinct classes, obtaining $H = K_{|V_1|,\dots,|V_k|}$. The coloring f remains proper on H, so w is still a legal disturbance. Every proper k-coloring of H is proper on G. This inclusion also holds after restricting to colorings with the same labeled counts as w. Thus the minimum repair cost for w cannot decrease, under either repair rule, and $F(G) \le F(H)$. The reverse inequality between the two maxima follows because the complete k-partite graphs are among the graphs on the left. □

5.1. Count-preserving repairs.

Theorem 5.2. *For $1 \le k \le n$, one has $\mathcal{M}(n,1) = 0$ and*

$$\mathcal{M}(n,2) = 2\left\lfloor \frac{n-1}{2} \right\rfloor \qquad (n \ge 2).$$

For $k \ge 3$, put $d = n - k$. Then

$$\mathcal{M}(n,k) = \begin{cases} n-1, & n = 4d-1 \text{ or } (n,k) \in \{(5,3),(7,3)\},\\ \min\{n, 4d\}, & \text{otherwise.}\end{cases}$$

The proof uses the criterion in Theorem 4.2: if $\mathbf{p} = (p_1, \dots, p_k)$ has total n, then

$$B(K_{\mathbf{p}}) = n \iff s\,|\{i : p_i = s\}| \le \frac{n}{2} \quad \text{for every occurring part size } s. \tag{5.1}$$

The relevant quantity is therefore the combined size of all parts of the same size, rather than the size of an individual part.

Proof of Theorem 5.2. We first establish the upper bound

$$\mathcal{M}(n,k) \le \min\{n, 4(n-k)\}. \tag{5.2}$$

By Theorem 5.1, it is enough to consider $K_{\mathbf{p}}$. Let a be its number of nonsingleton parts, and let T be their total size. Since

$$d = n - k = \sum_{i=1}^{k} (p_i - 1),$$

each nonsingleton part contributes at least one to d, whence $a \le d$ and $T = d + a \le 2d$.

Represent a disturbance by its matrix $M = (m_{ij})$, with row i corresponding to part i and column j to color j. As in (4.2), the corresponding row and column sums are p_i. The rows belonging to singleton parts have total mass $n - T$. At most T of this mass can lie in nonsingleton columns. Consequently the singleton-to-singleton block has mass at least $n - 2T \ge n - 4d$. Each positive entry in this block is 1, and no two positive entries share a row or a column. These entries extend to a permutation of the singleton indices. Completing the permutation within each of the other part-size classes gives an allowed repair retaining all those entries. Thus at most $4d$ vertices need to change, which proves (5.2).

Suppose first that $n \geq 4d$. Take d parts of size 2 and $n-2d$ singleton parts; their number is $n-d=k$. For each size-2 part, choose two distinct, previously unused singleton parts. Move the two tokens of its source color to these singletons, and move the two singleton colors into the size-2 part. There are enough singleton parts because $n-2d \geq 2d$. Leave the remaining singleton colors fixed. The $4d$ exchanged vertices carry colors whose multiplicities differ from the sizes of their parts, so none can be retained by a count-preserving repair. All remaining vertices can be retained. This gives equality in (5.2). The construction also includes $d=0$, when no exchanges are made.

Next suppose $n=4d-1$, so $k=3d-1$. In any part-size vector, if $a<d$, the singleton class has total size

$$k-a \geq (3d-1)-(d-1)=2d>n/2.$$

If $a=d$, then every nonsingleton part has size 2, and this part-size class has total size $2d>n/2$. In either case, (5.1) excludes villainy n, so $\mathcal{M}(n,k) \leq n-1$. To attain $n-1$, take d size-2 parts and $2d-1$ singletons. Perform the preceding exchanges for $d-1$ of the size-2 parts. In the last size-2 part, exchange one color token with the last singleton and leave the other token fixed. Exactly one unit of the disturbance matrix lies in blocks whose row and column parts have the same size. It is a diagonal entry, so the maximum number of retained vertices is exactly one.

It remains to find profiles satisfying (5.1) when $n<4d-1$, apart from the two stated small exceptions. The following construction also covers the boundary $n=4d$. Put

$$q=4d-n=3n-4k.$$

We claim that such a profile exists whenever $k \geq 3$, $q \geq 0$, and $q \neq 1$, except for $(n,k)=(5,3)$ and $(7,3)$.

If a profile satisfies (5.1), adjoining the three parts $(1,1,2)$ preserves the criterion. The order increases by 4, so the permitted mass $n/2$ increases by 2. The size-1 and size-2 classes each gain exactly 2, and all other classes are unchanged. It is therefore enough to use the seed profiles in the following table and repeatedly adjoin $(1,1,2)$.

Number of parts	Order	Part sizes
3	4	$(1,1,2)$
3	$2m \geq 6$	$(1,m-1,m)$
3	$2m+1 \geq 9$	$(2,m-1,m)$
4	$2m \geq 6$	$(1,1,m-2,m)$
4	$2m+1 \geq 7$	$(1,1,m-1,m)$
5	$2m \geq 8$	$(1,1,1,m-3,m)$
5	$2m+1 \geq 9$	$(1,1,1,m-2,m)$
6	9	$(1,1,1,1,2,3)$
6	11	$(1,1,1,1,2,5)$

Each seed satisfies (5.1). In the ranged families, the nonsingleton sizes are distinct and at most half the order. The only endpoints at which an additional displayed part becomes a singleton are $(1,1,1,3)$ and $(1,1,1,1,4)$; their singleton masses are also at most half the order. The largest equal-size class in the two six-part seeds has mass 4 and 5, respectively.

For completeness, these seeds cover all the claimed parameters. Choose $k_0 \in \{3,4,5\}$ with $k_0 \equiv k \pmod 3$, and set

$$j=\frac{k-k_0}{3}, \qquad n_0=n-4j.$$

Then $j \geq 0$ and $3n_0-4k_0=q$. For $k_0=4$, this gives $n_0 \geq 6$, so a four-part seed applies. For $k_0=5$, it gives $n_0 \geq 7$; equality would imply $q=1$, which is excluded, so a five-part seed applies. For $k_0=3$, it gives $n_0 \geq 4$, and a three-part seed applies unless $n_0 \in \{5,7\}$. In those two cases, if $j \geq 1$, use the six-part seed of order n_0+4 and adjoin $j-1$ copies of $(1,1,2)$. If $j=0$, the parameters are precisely the two excluded pairs. In every other case, adjoin j copies to the chosen k_0-part seed. This proves the claim, and hence yields $\mathcal{M}(n,k)=n$ throughout this range.

For $(n,k)=(5,3)$, the possible unordered part-size vectors are $(1,1,3)$ and $(1,2,2)$. For $(n,k)=(7,3)$, they are

$$(1,1,5), \quad (1,2,4), \quad (1,3,3), \quad (2,2,3).$$

Every one has an equal-size class of mass greater than $n/2$, so the extremal villainy is at most $n-1$. The disturbance matrices

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

have respective margin vectors $(1,1,3)$ and $(1,2,4)$. In each matrix, an allowed permutation retains at most one vertex, and the identity retains one. Both upper bounds are therefore attained.

Finally, $k=1$ gives zero villainy. For $k=2$, apply the complete bipartite formulas in Theorem 4.6. Unequal parts $a<b$ give $B(K_{a,b})=2a$, whereas equal parts give $B(K_{a,a})=2\lfloor a/2\rfloor$. Their maximum at fixed sum n is $2\lfloor (n-1)/2\rfloor$. It is attained by $((n-1)/2,(n+1)/2)$ for odd n, and by $(n/2-1,n/2+1)$ for even $n\geq 4$. When $n=2$, the graph is K_2 and the value is zero. □

5.2. **Weak repairs and an assignment parity obstruction.** Call a square matrix *balanced* if its ith row sum equals its ith column sum for every i. The row sums need not all be equal. Let $\mathcal{B}_{n,k}$ be the set of balanced, nonnegative integer $k\times k$ matrices with total sum n and with all row sums positive. Theorem 5.1 and (4.1) give

$$\mathcal{M}_{\mathrm{w}}(n,k)=n-\min_{M\in\mathcal{B}_{n,k}}\tau(M), \qquad \tau(M)=\max_{\sigma\in S_k}\sum_{i=1}^{k} m_{i,\sigma(i)}. \tag{5.3}$$

Theorem 5.3. *One has $\mathcal{M}_{\mathrm{w}}(n,1)=0$. For $2\leq k\leq n$, put*

$$L=\max\left\{\left\lceil\frac{n}{k}\right\rceil,\,2k-n\right\},$$

and define

$$\varepsilon=\begin{cases}1, & L \text{ is odd}, \quad (k \text{ is even or } L<k), \quad n\geq k(L-1)+2,\\ 0, & \text{otherwise.}\end{cases}$$

Then

$$\mathcal{M}_{\mathrm{w}}(n,k)=n-L-\varepsilon.$$

The maximum is attained by a complete k-partite graph with at most three distinct part sizes.

We prove the theorem after two assignment bounds. Besides averaging, integrality and balance impose the parity restriction responsible for ε.

Lemma 5.4. *If M is a nonnegative integer $k\times k$ matrix of total sum n with positive row and column sums, then*

$$\tau(M)\geq\max\left\{\left\lceil\frac{n}{k}\right\rceil,\,2k-n\right\}.$$

Proof. Each entry occurs in exactly $(k-1)!$ of the $k!$ permutation sums. Their average is consequently n/k, giving the first bound.

For the second, form the bipartite graph with one vertex for each row and each column, joining a row to a column when their entry is positive. This graph has $2k$ vertices, no isolated vertices, and at most n edges. If it has c components and e edges, then $e\geq 2k-c$, since each connected component on v vertices has at least $v-1$ edges. Hence $c\geq 2k-n$. Choose one edge from each component. These edges form a matching, and their entries have total weight at least c. Extend this matching to a permutation, using zero entries when necessary. Nonnegativity then gives $\tau(M)\geq c\geq 2k-n$. □

The elementary bounds use only positivity of the margins. Balance adds a restriction when the assignment score is close to its average lower bound: the two sets of optimal potentials must then differ by a constant. This is the source of the parity correction.

Lemma 5.5 (Balanced assignment bound)**.** *Let M be a balanced, nonnegative integer matrix of order $k\geq 2$, with total sum n and assignment score $t=\tau(M)$. If*

$$n>k(t-1)+1,$$

then there exist $\mathbf{a}\in\mathbb{Z}_{\geq 0}^{k}$ and $c\in\{0,1\}$ such that

$$m_{ij}\leq a_i+a_j+c \quad (i,j\in[k]), \qquad t=2\sum_i a_i+kc. \tag{5.4}$$

Consequently, if t is odd and either k is even or $t<k$, then

$$n\leq k(t-1)+1.$$

Proof. Take nonnegative integer optimal potentials $\mathbf{u}, \mathbf{v}$ from Theorem 4.1, retaining the original column labels. Thus $m_{ij} \le u_i + v_j$ and $\sum_i (u_i + v_i) = t$. Set

$$z_{ij} = u_i + v_j - m_{ij}, \qquad d_i = u_i - v_i, \qquad \eta = \sum_i d_i.$$

Nonnegativity and balance give

$$\Delta := \sum_{i,j} z_{ij} = kt - n, \qquad \sum_j z_{ij} - \sum_j z_{ji} = kd_i - \eta. \tag{5.5}$$

Suppose that the integers d_i are not all equal. Let $b = \min_i d_i$, $I = \{i : d_i = b\}$, $q = |I|$, and $S = \sum_i (d_i - b)$. Then $1 \le q \le k-1$ and $S \ge k - q$. Since $\eta = kb + S$, summing the second identity in (5.5) over I and cancelling internal terms gives

$$\sum_{\substack{i \in I \\ j \notin I}} z_{ji} - \sum_{\substack{i \in I \\ j \notin I}} z_{ij} = qS.$$

It follows that

$$\Delta \ge qS \ge q(k-q) \ge k - 1,$$

where the last inequality is equivalent to $(q-1)(k-q-1) \ge 0$.

Under the stated hypothesis, $\Delta < k - 1$, so all d_i equal an integer d. Put

$$a_i = \min\{u_i, v_i\} + \left\lfloor \frac{|d|}{2} \right\rfloor, \qquad c = |d| - 2\left\lfloor \frac{|d|}{2} \right\rfloor.$$

These quantities are nonnegative integers and $c \in \{0, 1\}$. Because $u_i - v_i = d$ for every i, we have

$$u_i + v_j = a_i + a_j + c.$$

Summing the corresponding diagonal identity gives $t = 2\sum_i a_i + kc$, proving (5.4). If t is odd, this identity forces $c = 1$ and k odd, and hence $t \ge k$. Its contrapositive gives the final assertion. □

Proof of Theorem 5.3. The case $k = 1$ is immediate. Assume $k \ge 2$ and write $t = L + \varepsilon$. Theorem 5.4 gives $\tau(M) \ge L$ for every $M \in \mathcal{B}_{n,k}$. If $\varepsilon = 1$, Theorem 5.5 rules out $\tau(M) = L$. Thus

$$\tau(M) \ge t \qquad (M \in \mathcal{B}_{n,k}). \tag{5.6}$$

We construct a matrix attaining equality. The constructions are symmetric, which ensures balance automatically.

First suppose $k \le n \le 2k - 2$, and put $r = n - k$. On $r + 2$ indices, take the adjacency matrix of a star with $r + 1$ leaves. On each of the remaining $k - r - 2$ indices, put a diagonal entry 1; set every other entry to zero. The total sum is

$$2(r + 1) + (k - r - 2) = n,$$

and every row sum is positive. A permutation can use at most two positive entries from the star: at most one lies in the center row and at most one in the center column. Two are attainable by exchanging the center with a leaf. Fixing all the remaining indices therefore gives

$$\tau(M) = 2 + (k - r - 2) = 2k - n = L.$$

Here $\varepsilon = 0$: if L is odd, then $L \ge 3$, and $n \le 2k - 2 < k(L-1) + 2$. Hence this matrix has the required weight t.

For the remainder of the proof, assume $n \ge 2k - 1$. Then $L = \lceil n/k \rceil \ge 2$.

Suppose next that $t = 2h$ is even. Use index 1 as the center of a weighted star and set

$$m_{1i} = m_{i1} = a_i \quad (2 \le i \le k), \qquad m_{11} = b,$$

where $1 \le a_i \le h$ and $0 \le b \le 2h$; all other entries are zero. The sum $S = \sum_{i=2}^{k} a_i$ can be any integer between $k - 1$ and $h(k - 1)$. For each such S, the possible total sums $2S + b$ fill the integer interval $[2S, 2S + 2h]$. Consecutive intervals overlap, because $h \ge 1$. Together they cover every integer from $2k - 2$ to $2hk$. As $2k - 1 \le n \le kt = 2hk$, the entries can be chosen to give total sum n. For the chosen S, take the leaf weights as equally as possible, so that each is either $\lfloor S/(k-1) \rfloor$ or $\lceil S/(k-1) \rceil$; both values that occur lie between 1 and h. Every row sum is positive. The potentials

$$u_1 = v_1 = h, \qquad u_i = v_i = 0 \quad (i > 1)$$

satisfy $m_{ij} \le u_i + v_j$, so every permutation has weight at most $2h = t$. Equality follows from (5.6).

It remains to consider odd t. The definition of ε then forces $\varepsilon = 0$ and $t = L$. If k is even or $t < k$, the fact that $\varepsilon = 0$ implies $n \le k(t-1)+1$. On the other hand, $t = \lceil n/k \rceil$ implies $n > k(t-1)$. Thus

$$n = k(t-1)+1.$$

Write $t = 2h+1$, with $h \ge 1$. In the weighted-star construction, take every $a_i = h$ and take $b = t$. Its total sum is

$$2h(k-1) + (2h+1) = k(t-1)+1 = n.$$

The identity permutation has weight t. The potentials $u_1 = h+1$, $v_1 = h$, with all other potentials zero, give an upper bound of t, so $\tau(M) = t$.

Finally, suppose both t and k are odd and $t \ge k$. In particular, $k \ge 3$. The additive form in Theorem 5.5 suggests starting with a matrix of total sum kt whose assignment weights all equal t. Set

$$a_1 = \frac{t-k}{2}, \qquad a_i = 0 \quad (i > 1), \qquad m_{ij} = 1 + a_i + a_j.$$

This symmetric matrix has total sum kt, and every permutation has weight

$$k + 2\sum_i a_i = t.$$

Put $\delta = kt - n$. Since $t = \lceil n/k \rceil$, we have $0 \le \delta \le k-1$. Choose δ indices from $\{2, \dots, k\}$ and reduce their diagonal entries from 1 to 0. The resulting matrix remains symmetric, has positive row sums, and has total sum n. No assignment weight increases. A cyclic permutation of the k indices avoids every altered diagonal entry and still has weight t. This gives the desired matrix in the last case.

Each constructed matrix belongs to $\mathcal{B}_{n,k}$ and hence represents a legal disturbance on the complete multipartite graph whose part sizes are its row sums. Equation (5.3) and (5.6) now prove the formula.

These extremal graphs need at most three distinct part sizes. The first and third constructions have at most two distinct row sums. In the even weighted-star construction, the leaf weights take at most two consecutive values, and the center supplies at most one further row sum. In the final construction, writing $a = (t-k)/2$, the row sums are $k + (k+1)a$ at the first index and either $k+a$ or $k+a-1$ at the other indices. Thus there are at most three row sums in every case. □

6. Concluding remarks

Prescribed multiplicities and agreement quotas can be met simultaneously on odd cycles, while complete multipartite repairs reduce to integral assignments. Optimizing the latter description over the part sizes determines the extrema over all graphs of fixed order and chromatic number. The remaining questions concern which parts of these descriptions extend beyond the settings treated here.

Problem 6.1. Beyond the odd-cycle case of Theorem 3.8, characterize graph–palette pairs for which every extendable nonempty partial coloring admits every integer profile coordinatewise between two of its extension profiles, subject to the fixed total. For graphs of bounded treewidth and a fixed palette, determine sharp simultaneous per-color agreement quotas when a feasible target profile is prescribed.

The second question concerns guaranteed agreement counts, in addition to existence of a coloring with the prescribed profile. For comparison, Bradshaw, Masařík, and Stacho [3] prove that graphs of treewidth two are 1/3-flexibly three-choosable, without prescribing the target multiplicities. This means that, for arbitrary three-element lists of allowed colors and a request for one listed color at each vertex of any subset R, some proper coloring from the lists satisfies at least $|R|/3$ requests.

The ordinary multipartite formula has at most k^2 candidates. For weak villainy, the ordered-potential search depends on the spread of the part sizes, and Theorem 4.9 gives a polynomial-time algorithm for every fixed number of parts.

Question 6.2. Can $B_{\mathrm{w}}(K_{p_1,\dots,p_k})$ be computed in time polynomial in the binary encoding length of $(p_1, \dots, p_k)$, when the number of parts is also part of the input?

Theorem 5.1 guarantees a complete multipartite extremizer, but need not capture all graphs attaining the same value.

Problem 6.3. Characterize the graphs attaining $\mathcal{M}(n,k)$ and $\mathcal{M}_{\mathrm{w}}(n,k)$. In particular, determine which edges can be removed from a complete multipartite extremizer while preserving both its chromatic number and its extremal villainy.

One can also let $B_s(G)$ denote the worst repair cost when the target profile may differ from the source profile by at most $2s$ in ℓ_1 distance. Then $B_0(G) = B(G)$, and $B_s(G) = B_{\mathrm{w}}(G)$ as soon as

$$s \geq \min\{B_{\mathrm{w}}(G),\, |V(G)| - \chi(G)\}.$$

Indeed, changing d vertices changes the profile by at most $2d$, and two positive k-color profiles on n vertices have ℓ_1 distance at most $2(n-k)$. For odd cycles, Theorem 3.15 already gives equality for every $s \geq 0$, even with the source profile fixed. When $B(G) > B_{\mathrm{w}}(G)$, determining the intermediate values and the least profile freedom needed to attain the weak value remains a further direction.

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The authors used AI tools from OpenAI to assist with developing and checking arguments, computational experiments, literature searches, and manuscript preparation. The authors are responsible for all proofs, computations, and conclusions.

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